ELEC5305 Project Proposal

Hybrid Multichannel Speech Enhancement Using MVDR Beamforming: A Controlled Comparison of Wiener Post-Filter Frequency Smoothing

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1. Student Information

Item	Details
Full Name	Peinan Gu
Student ID (SID)	530679835
GitHub Username	kira8888-bb
GitHub Repository	https://github.com/kira8888-bb/elec5305-project-530679835
GitHub Pages	https://kira8888-bb.github.io/elec5305-project-530679835/

After deployment, the GitHub Pages URL will be opened in a signed-out browser to confirm public access. Until that check passes, the repository README remains the primary public project summary.

2. Project Overview

Speech recordings suffer from background noise, competing speakers and room reverberation. This project investigates an established hybrid architecture combining frequency-domain Minimum Variance Distortionless Response (MVDR) beamforming with Wiener post-filtering. It does not claim that adding auditory-band Wiener processing is new.

The main research question is:

> *Does ERB-band smoothing provide an advantage beyond conventional frequency-bin or uniformly grouped Wiener post-filtering after MVDR beamforming, and under what input-SNR and reverberation conditions?*

A MATLAB simulation will isolate frequency grouping while holding the spatial processing and power-estimation method constant. The intended contribution is a reproducible comparison explaining the trade-off between estimate stability and spectral-detail preservation.

3. Background and Motivation

Capon beamforming [1] and multichannel Wiener filtering [2] provide the spatial-filtering foundation. Their combination with post-filtering is established. Gui and Kwan [3] already investigated adaptive Wiener filtering in critical-band gammatone subbands. Li et al. [4] studied microphone-array post-filtering using auditory filtering. These closely related studies motivate replication and comparison rather than an architectural novelty claim.

The ERB scale reflects frequency-dependent auditory-filter bandwidths [5]. Pooling neighbouring frequency bins may reduce estimation variance, but ordinary frequency averaging can also do this. Wider grouping may blur harmonics or over-attenuate speech. Therefore, an ERB advantage must be assessed against both ungrouped and uniform-frequency controls; superiority is not assumed.

Neural mask estimation [7] and learned spatial-covariance estimation [8] place this classical system in a modern context. No state-of-the-art performance claim or neural-network training is planned. The classical approach remains useful because its spatial and spectral operations are interpretable and computationally modest.

4. Proposed Methodology

Controlled simulation. Fix a four-microphone linear array, room dimensions, target position and interfering-source position. Use an established image-source simulator [6], initially MATLAB acousticRoomResponse. The microphone signals are

$$x_m[n] = h_{s,m}[n] * s[n] + h_{i,m}[n] * i[n] + v_m[n]. \quad (1)$$

Test two interference conditions (stationary directional noise and competing speech), three input SNRs {-5, 0, 5} dB, and two nominal RT60 values {0.2, 0.6} s: twelve conditions. Record achieved decay estimates. Use multiple speech utterances and repeated seeds, with identical mixtures across methods. Synthetic signals are reserved for engineering tests.

Spatial processing. Apply STFT, delay-and-sum and diagonally loaded MVDR, followed by inverse STFT. Use known direct-path geometry initially; any known full-RIR steering experiment is a separately labelled diagnostic. Cross-check the standard MVDR solution against an established implementation/example and test reconstruction and distortionless constraints before interpreting results.

Shared estimator. An explicit target-absent calibration segment estimates interference-plus-noise covariance; the beamformer and noise PSD are then held fixed. At the MVDR output, noise power is w^H R_n w. Recursively smooth mixture power with alpha = 0.8 and estimate speech power by nonnegative subtraction:

$$Q_{k,l} = \alpha Q_{k,l-1} + (1-\alpha)|Y_{k,l}|^2, \quad P_{s,k,l} = \max(Q_{k,l} - P_{n,k}, 0). \quad (2)$$

This initial estimator is not decision-directed and assumes representative calibration noise. All variants share these estimates, initialization, STFT settings and gain floor. Nonstationary interference exposes this assumption; later estimator changes must apply to every variant.

Frequency grouping. Compare identity/bin-wise processing, uniform-Hz bands, and ERB-spaced bands. Both grouped systems use 24 overlapping triangular weights, linear in Hz, covering DC to Nyquist, with identical normalization and interpolation rules. ERB centres are equally spaced on E(f) = 21.4 log10(1 + 0.00437f) [5]. Average speech and noise powers separately within each band, then calculate

$$\xi_{b,l} = P_{s,b,l} / \max(P_{n,b}, \varepsilon), \quad G_{b,l} = \xi_{b,l} / (1 + \xi_{b,l}). \quad (3)$$

Map gains back to STFT bins and apply a common 0.05 floor. This is cochlear-inspired frequency smoothing, not a complete cochlear or gammatone model. Check sensitivity to 16/24/32 bands on development data; equal band counts do not equalize every local bandwidth.

Compare seven systems: unprocessed reference microphone; single-channel Wiener; delay-and-sum; MVDR; and MVDR plus bin-wise, uniform-band or ERB-band Wiener. The last three are the primary comparison.

Evaluation. Use the desired reverberant target component at the reference microphone, s_ref, with common timing and no output-specific normalization. Define reference-error SNR and its improvement explicitly:

$$\mathrm{SNR}_e(y) = 10 \log_{10}(\|s_{ref}\|^2 / \|s_{ref}-y\|^2),$$
$$\Delta\mathrm{SNR}_e = \mathrm{SNR}_e(y) - \mathrm{SNR}_e(x_{ref}). \quad (4)$$

This includes speech distortion, unlike a pure output noise-power ratio. Also report SI-SDR [9] and STOI [10]; with a reverberant reference, STOI describes preservation relative to that reference. Analyse paired utterance-level differences and uncertainty, across-seed gain variance, spectrograms and listening examples. Lower gain variance alone does not imply better speech quality.

5. Expected Outcomes

The outcome will be a reproducible MATLAB prototype and evidence showing whether auditory frequency organization adds value beyond generic smoothing. Low-SNR stability benefits and high-SNR detail loss are hypotheses, not promised results. Null or negative ERB findings remain informative.

Deliverables include source code, configurations, quantitative comparisons, figures, permitted audio examples, a report and video. The README and verified GitHub Pages site will explain reproduction, prior work, calibration assumptions and limitations. Angular-separation experiments remain optional.

6. Timeline

The original week-based structure is retained as a planned sequence, not a completion record.

Weeks	Task
Weeks 1-3	Review literature, revise research positioning and verify GitHub Pages.
Weeks 4-5	Reuse room/STFT tools; validate geometry, mixtures and reverberant reference.
Weeks 6-7	Validate MVDR and establish the shared PSD estimator and bin-wise baseline.
Weeks 8-9	Add uniform and ERB grouping; check fairness and development parameters.
Weeks 10-11	Run the twelve-condition paired speech experiments and stability analysis.
Week 12	Interpret scores, distortion, spectrograms and listening examples.
Week 13	Complete report, video, reproducibility documentation and website checks.

7. References

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